# Standard Operating Procedure for Resolving Major Optical Transport Failures on Arista Access Nodes at Hebbal Edge

**Node:** HEB-ACC-11 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8236

## Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Arista Access Nodes at Hebbal Edge is a critical failure domain that can have a significant impact on network availability and reliability. The SFP/XFP modules used on these nodes are responsible for transmitting and receiving optical signals over fiber cables. If these modules fail or become degraded, it can lead to a loss of connectivity between the node and the upstream/downstream network elements. This failure can be further exacerbated by fiber cuts or Rx/Tx power loss issues, which can prevent the node from transmitting or receiving signals. The Arista Access Nodes at Hebbal Edge use the eAPI (Extensible API) to manage and monitor the SFP/XFP modules, which allows for real-time monitoring and control of these critical network elements. In the event of an Optical Transport Failure, it is essential to quickly identify and resolve the root cause to minimize network downtime and ensure high availability.

## Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible and the node is powered on
- Check the redundant link status to ensure that the node is not relying solely on the failed link
- Dispatch remote hands to the Hebbal Edge site to assist with physical inspections and repairs

## Potential Root Causes

| Cause | Probability |
|---|---|
| SFP/XFP Module Failure | High |
| Fiber Cut | Medium |
| Rx/Tx Power Loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show interfaces e1/1/0 detail, Expected Output: Interface e1/1/0 is up, line protocol is up, Description: This command is used to verify the status of the SFP/XFP module and the link it is connected to. The expected output indicates that the interface is up and the line protocol is up, which suggests that the SFP/XFP module is functioning correctly.

Command: show interfaces e1/1/0 counters, Expected Output: Interface e1/1/0 has received 1000 packets and transmitted 1000 packets, Description: This command is used to verify the packet count and error
```

```
statistics for the SFP/XFP module. The expected output indicates that the interface has received and transmitted a significant number of packets, which suggests that the SFP/XFP module is functioning correctly.

Command: show interface e1/1/0 transceiver, Expected Output: Transceiver type is SFP, transceiver state is up, Description: This command is used to verify the type and state of the SFP/XFP module. The expected output indicates that the transceiver type is SFP and the transceiver state is up, which suggests that the SFP/XFP module is functioning correctly.
```

## Resolution Steps

Step 1: Step: Replace the failed SFP/XFP module, Command: no interface e1/1/0, Description: This command is used to shut down the interface and remove the failed SFP/XFP module. Once the module is removed, the new module can be inserted and the interface can be brought back up using the command 'interface e1/1/0'.

Step 2: Step: Verify the link status and packet count, Command: show interfaces e1/1/0 detail, Description: This command is used to verify the status of the SFP/XFP module and the link it is connected to. The expected output indicates that the interface is up and the line protocol is up, which suggests that the SFP/XFP module is functioning correctly.

Step 3: Step: Verify the packet count and error statistics, Command: show interfaces e1/1/0 counters, Description: This command is used to verify the packet count and error statistics for the SFP/XFP module. The expected output indicates that the interface has received and transmitted a significant number of packets, which suggests that the SFP/XFP module is functioning correctly.

## Post-Check Verification

```
Command: show interfaces e1/1/0 detail, Expected Output: Interface e1/1/0 is up, line protocol is up, Description: This command is used to verify the status of the SFP/XFP module and the link it is connected to. The expected output indicates that the interface is up and the line protocol is up, which suggests that the SFP/XFP module is functioning correctly.

Command: show interfaces e1/1/0 counters, Expected Output: Interface e1/1/0 has received 1000 packets and transmitted 1000 packets, Description: This command is used to verify the packet count and error statistics for the SFP/XFP module. The expected output indicates that the interface has received and transmitted a significant number of packets, which suggests that the SFP/XFP module is functioning correctly.
```

## Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: interface e1/1/0, Description: This command is used to restore the original SFP/XFP module that was removed during the resolution process. Once the module is restored, the interface can be brought back up using the command 'no shutdown'.

- Step: Verify the link status and packet count, Command: show interfaces e1/1/0 detail, Description: This command is used to verify the status of the SFP/XFP module and the link it is connected to. The expected output indicates that the interface is up and the line protocol is up, which suggests that the SFP/XFP module is functioning correctly.